BrightLight Tutorial - Data Analytics

SQL Fundamentals

Exercise 1

id	first_name	last_name	department	salary	hiredate	city
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	53000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

1. SELECT statement

```
SELECT *
FROM employees;
```

2. DISTINCT

```
SELECT DISTINCT department
FROM employees;
```

3. ORDER

```
SELECT first_name, last_name
FROM employees
ORDER BY salary DESC;
```

4. LIMIT

```
SELECT *
FROM employees
ORDER BY salary DESC
LIMIT 5;
```

5. WHERE

```
SELECT *
FROM employees
WHERE department = 'IT';
```

6. AND

```
SELECT *
FROM employees
WHERE department = 'Finance'
  AND salary > 58000;
```

7. OR

```
SELECT *
FROM employees
WHERE department = 'HR'
OR department = 'Marketing'
```

8. NOT

```
SELECT *
FROM employees
WHERE department <> 'IT';
```

9.

9. IN

```
SELECT *
FROM employees
WHERE department IN('HR', 'IT', 'Finance');
```

10. SELECT *

```
FROM employees
WHERE department = 'IT'
AND salary > 50000
AND city = 'New York';
```

11. Combining WHERE, AND, and ORDER BY

```
SELECT first_name, last_name
FROM employees
WHERE department IN('Finance', 'Marketing')
AND salary > 52000
ORDER BY Salary DESC
```

12. Combining SELECT DISTINCT, WHERE, and IN

```
SELECT DISTINCT city
FROM employees
WHERE department NOT IN('IT', 'HR');
```

13. Combining WHERE, NOT, AND, and ORDER BY

```
SELECT *
FROM employees
WHERE department <> 'Finance' AND salary > 50000
ORDER BY hire_date ASC;
```

14. SELECT *
FROM employees
WHERE city IN('chicago', 'Los Angeles')
AND department IN ('IT', 'Marketing')
LIMIT 3;